

- 2 A student observes the orbits of some of the moons around the planet Saturn, as shown in Fig. 2.1.

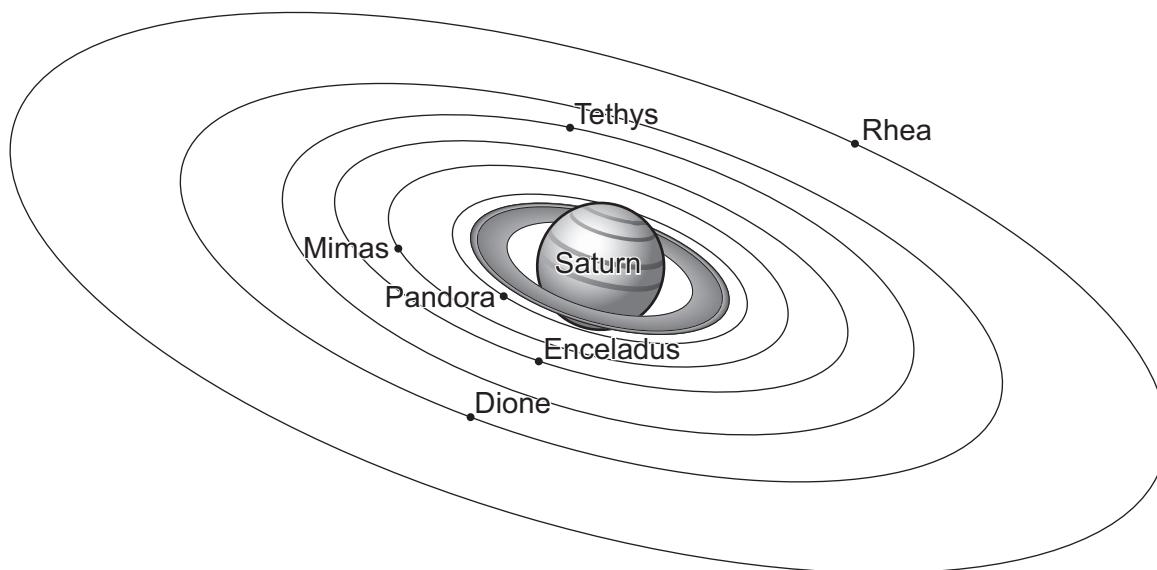


Fig. 2.1 (not to scale)

For the moon Pandora, the period of the orbit and the mean distance from the centre of Saturn are determined.

The measurements of period T and mean distance r are repeated for other moons.

It is suggested that T and r are related by the equation

$$T = \frac{2\pi r^n}{k}$$

where n and k are constants.

- (a) A graph is plotted of $\lg T$ on the y -axis against $\lg r$ on the x -axis.

Determine expressions for the gradient and y -intercept.

gradient =

y -intercept =

[1]

